DR. A. P. J. ABDUL KALAM TECHNICAL UNIVERSITY

LUCKNOW, UTTAR PRADESH

STUDY & EVALUATION SCHEME WITH SYLLABUS

FOR

B. TECH. 4th YEAR

AERONAUTICAL ENGINEERING

[Effective from Session: 2021-22]

B. Tech Aeronautical Engineering
Evaluation Scheme
Effective from Session 2021-22

Semester VII											
S. No.	Subject Code	Subject Name	Periods	Evaluation Scheme				End Semester		Total	Credit
			L-T-P	CT	TA	Total	PS	TE	PE		
1		HSMC-1/ HSMC-2	3-0-0	30	20	50		100		150	3
2		Department Elective-IV	3-0-0	30	20	50		100		150	3
3		Department Elective-V	3-0-0	30	20	50		100		150	3
4		Open Elective-II	3-0-0	30	20	50		100		150	3
5	KAE 751	Aircraft Systems & Maintenance Lab	0-0-2				25		25	50	1
6	KAE 752	Mini Project/ (Internal Assessment*)	0-0-2				50			50	1
7	KAE 753	Project	0-0-8				150			150	4
8		MOOCs (Essential for Honors Degree)									
Total			12-0-12							850	18

Semester VIII											
S. No.	Subject Code	Subject Name	Periods	Evaluation Scheme				End Semester		Total	Credit
			L-T-P	CT	TA	Total	PS	TE	PE		
1		HSMC-2 / HSMC-1	3-0-0	30	20	50		100		150	3
2		Open Elective-III	3-0-0	30	20	50		100		150	3
3		Open Elective-IV	3-0-0	30	20	50		100		150	3
4	KAE 851	Project	0-0-18				100		300	400	9
5		MOOCs (Essential for Honors Degree)									
Total			9-0-18							850	18

Department Electives-IV	Department Electives-V
1. **KAE-071 :** Flight Mechanics	1. **KAE-075 :** Aircraft Rules & Regulation
2. **KAE-072 :** Aircraft Stability & Control	2. **KAE-076 :** Aircraft Maintenance
3. **KAE-073 :** Helicopter Aerodynamics	3. **KAE-077 :** Introduction to CFD
4. **KAE-074 :** Introduction to Finite Element Methods	4. **KAE-078 :** Elements of Aero-elasticity

Subject Code:KAE-771	Aircraft System and Maintenance Lab	L T P: 0 0 2	Credits:1

Course Outcome: The students will be able to		Blooms Taxonomy
CO-1	Understand the operation of radio receivers in the frequency band of HF and VHF, direction finding	K2
CO-2	Understand the function of pressure operated flight instruments and their function.	K2
CO-3	Apply the concept gyro to understand the operation of gyro horizon to visualize attitude of aircraft.	K2
CO-4	Understand the landing gear operation and power plant indication in cockpit instrument panel.	K3
CO-5	Apply the concepts of Hydraulic and pneumatic system to operate landing gear and flight controls.	K3

Minimum 10 out of following such Experiments (or such experiments)

1. Familiarization with various communication systems such as High Frequency receiver, Very High Frequency receiver.
2. Familiarization with Automatic Direction-finding system installation and working.
3. Familiarization with VHF Omni Range/Instrument Landing systems.
4. Familiarization with Artificial horizons. Slip indicators and Directional gyros
5. Familiarization with pressure devices such as Pitot static system, Altimeters. Vertical speed indicator.
6. Familiarization with direct reading temperature indicating system.
7. Familiarization with Magnetic compass and direct reading compass.
8. Familiarization with Fuel quantity indicating system.
9. Familiarization with Pressurization System of aircraft.
10. Functional Test of aircraft landing gear retraction system and its relevant indications in the cockpit.
11. Identification of various components, pipelines with color coding on aircraft.
12. Study of hydraulic systems and its various components used in aircraft.
13. Study of pneumatic systems and its various components used in aircraft.
14. Study of brake systems of aircraft.

Department Electives-IV

Subject Code: KAE-071	Flight Mechanics	L T P: 3 0 0	Credits: 3

After completion of this course, the students will be able to		Blooms Taxonomy
CO-1	Understand the standard atmosphere and its significance.	K2
CO-2	Understand the performance of an aircraft during different cruising and maneuvering regimes of flight.	K2
CO-3	Understand the conditions corresponding to static and dynamic stability of an aircraft.	K2
CO-4	Apply the principles of dynamic stability to prevent an aircraft from different divergent and oscillatory modes.	K3
CO-5	Apply the knowledge of structural, propulsion and aerodynamic considerations to predict the safe operating velocity for an aircraft.	K3

UNIT-1 **(8 hours)**

Introduction: Standard Atmosphere, variation of pressure and density with altitude, geo-potential and geometric altitude, pressure, temperature and density altitude. Forces and moments acting on a flight vehicle - Equation of motion of a rigid flight vehicle, Different types of drag, Drag polar, Types of airspeeds, airspeed indicator, altimeter, rate of climb indicator, mach meter, Variation of thrust and power with velocity and altitude for air breathing engines.

UNIT-2 **(8 hours)**

Cruising and Maneuvering Flight Performance: Performance of airplane in level flight, power available and power required curves. Maximum speed in level flight, Conditions for minimum drag and power required, Range and endurance of aircraft, Climbing and gliding flight (Maximum rate of climb and steepest angle of climb, minimum rate of sink and shallowest angle of glide), Turning rate and turn radius, Bank rate and bank angle, load factor, limitations on turn, V-n diagram, load factor and Gust envelope, take-off and landing performance.

UNIT-3 **(8 hours)**

Static Longitudinal Stability: Static and dynamic stability - Purpose of controls in airplanes – Static, Longitudinal stability-Stick fixed stability, Basic equilibrium equation-Stability criterion-Effects of fuselage and tail; Influence of CG location- Power effects - Stick fixed neutral point and static margin- Stick free stability-Hinge moment coefficient -Stick free neutral points- Stick force gradients

UNIT-4 **(8 hours)**

Static Lateral and Directional Stability: Dihedral effect - Lateral control - Coupling between rolling and yawing moments – Roll stability and control, Adverse yaw effects- Aileron reversal - Static directional stability and control, Weathercock effect - Rudder requirements, Rudder lock. Aerodynamic balancing, peddle fixed and peddle stability, roll control reversal.

UNIT-5 **(8 hours)**

Dynamic Stability : Degree of freedom of rigid bodies in space, Introduction to dynamic

longitudinal stability: - Euler angles, velocities in 3- direction, Decoupling of longitudinal and lateral-directional dynamics; Modes of stability- longitudinal modes; lateral-directional modes, effect of freeing the stick- Brief description of lateral and directional Dynamic stability: Phugoid motion, short period motion, Spiral, divergence, Dutch roll, auto rotation and spin, approximation of longitudinal and lateral modes, Factors affecting time period and damping.

Book and References:

1. Perkins, C.D., and Hage, R.E., "Airplane Performance stability and Control", John Wiley
2. Nelson, R.C. "Flight Stability and Automatic Control", McGraw-Hill Book Co
3. Mc Cornick W., "Aerodynamics, Aeronautics and Flight Mechanics", John Wiley
4. Babister, A.W., "Aircraft Dynamic Stability and Response", Pergamon Press, Oxford
5. J.D, Anderson, "Introduction to Flight", Tata McGraw Hill

Subject Code: KAE-072	Aircraft Stability and Control	L T P: 3 0 0	Credits: 3

After completion of this course, the students will be able to		Blooms Taxonomy
CO-1	Understand the basics of control theory and its application in aircraft control systems.	K2
CO-2	Understand the PI, PD and PID controller design and its implementation in avionics components.	K2
CO-3	Understand the functioning of auto-pilot system in an aircraft.	K2
CO-4	Apply the concept of aircraft control for obstacle clearance and motion planning	K3
CO-5	Apply the stability criteria to determine the stability level for different configurations of aircraft.	K3

UNIT-1 **(8 hours)**

Introduction to Modern Control Theory, Introduction to state-space versus transform methods in linear systems, internal versus input/output formulation, discrete-time and continuous- time systems, Fundamental concepts of linearity, causality, time-invariance, Solution to LTI and LTV systems, Solutions to homogeneous and non-homogeneous cases. Computation of matrix exponentials using Laplace transforms and Jordan Normal form, positive definite matrices, quadratic forms.

UNIT-2 **(8 hours)**

Coordinate systems, Attitude dynamics and control, Rotational kinematics, Direction cosine matrix, Euler angles, Euler's Eigen axis rotation, Quaternion, Rigid body dynamics of launch vehicle, Angular momentum, Inertia matrix, Principal axes, Effect of aerodynamics, Generalized equations of motion, derivation of dynamic equations, structural dynamics and flexibility, actuator dynamics, gimbaled engine dynamics, External forces and moments, Linear model for Aero-structure-control-slosh interaction studies.

UNIT-3 **(8 hours)**

Classical linear time invariant control systems, Transfer function representations; stability; time domain characteristics, proportional–integral–derivative (PID) controller design for aerospace systems, Proportional term, Steady-state error, Integral term, Derivative term, Transient Response via Gain Adjustment, Lag Compensation, Lead Compensation, Lag-Lead Compensation.

UNIT-4 **(8 hours)**

Equations of motion, Stability derivatives, Routh's discriminant, solving the stability quartic, Stability definitions, characteristic equation, location of roots in the s-plane for stability, Routh-Hurwitz criteria of stability, Root locus and Bode techniques and its application to controller design for aerospace systems., concept and construction, frequency response. Gain Margin and Phase Margin via the Nyquist Diagram, Stability, Gain Margin, and Phase Margin via Bode Plots,

Relation Between Closed-Loop Transient and Closed-Loop Frequency Responses.

UNIT-5 **(8 hours)**

Basics of feedback control: History and motivation for feedback; terminologies, Frequency response, Stability concepts, Bandwidth, Transient response, Closed loop design specifications, tracking and disturbance rejection, Sensitivity to parameter variations. Obstacle avoidance and Motion planning, the canonical problem, Configuration space, Different types of obstacles, Planning via retraction, Planning via cell decomposition, Probabilistic planning, Planning via artificial potentials, Motion planning for manipulators.

Books and References:

1. Perkins C.D, & Hage, R.E. "Airplane Performance, Stability and Control", Wiley Toppan
2. Babister, A.W. "Aircraft stability and response ", Pergamum Press
3. Nelson, R.C. "Flight Stability & Automatic Control ", McGraw Hill

Subject Code: KAE-073	Helicopter Aerodynamics	L T P: 3 0 0	Credits: 3

After completion of this course, the students will be able to		Blooms Taxonomy
CO-1	Understand the functioning of a typical helicopter including the working of its flight control systems.	K2
CO-2	Understand the differences between the aerodynamics of an aircraft and a helicopter.	K2
CO-3	Understand the structural considerations of a helicopter including the study of failure modes.	K2
CO-4	Understand the flight performance of a helicopter during both hovering and forward flight.	K2
CO-5	Apply the above concepts to do preliminary design of a helicopter system.	K3

UNIT-1 **(8 hours)**

Types of helicopter, mechanism, characteristics, working of helicopter, Theory of Flight—Rotary Wing Aerodynamics: Terminology; Effects of gyroscopic precession; Torque reaction and directional control, Dissymmetry of lift, Blade tip stall; Translating tendency and its correction; Coriolis Effect and compensation; Vortex ring state, power settling.

UNIT-2 **(8 hours)**

Flight Control Systems: Cyclic control; Collective control; Swash plate; Yaw control: Anti-Torque Control, Tail rotor, bleed air; Main Rotor Head: Design and Operation features; Blade Dampers: Function and construction; Rotor Blades: Main and tail rotor blade construction and attachment; Trim control, fixed and adjustable stabilizers hovering performance.

UNIT-3 **(8 hours)**

Blade forces and motion in forward flight, Force, torque and flapping coefficient, Helicopter trim, Analysis, Performance in forward flight. Transmissions: Gear boxes, main and tail rotors; Clutches, free wheel units and rotor brake, Tail rotor drive shafts, flexible couplings, bearings, vibration dampers and bearing hangers.

UNIT-4 **(8 hours)**

Helicopter Structures: Airworthiness requirements for structural strength; Structural classification, primary, secondary and tertiary, fail safe, safe life, damage tolerance concepts; Zonal and station identification systems; Stress, strain, bending, compression, shear, torsion, tension, hoop stress, fatigue; Drain sand ventilation provisions.

UNIT-5 **(8 hours)**

Momentum and simple blade element theories, Blade advance ratio, Figure of merit-Profile and induced power estimation, Constant chord and ideal twist rotors, over pitching. Auto rotation and Ground effect, Rotor alignment; Main and tail rotor tracking; Static and dynamic balancing; Vibration types, vibration reduction methods; Ground resonance, helicopter spin and stall.

Books and Reference:

1. Automatic Flight Control – E.H.J. Pallet.
2. Aviation Maintenance Technician Handbook (General) 9A –FAA.
3. Helicopter Theory by Wayne Johnson.
4. Helicopter Calculation & Design Vol. I ,II & III by A.V. Nekrasov, A.S. Braverman.

Subject Code: KAE-074	Introduction to Finite Element Methods	L T P: 3 0 0	Credits: 3

After completion of this course, the students will be able to		Blooms Taxonomy
CO-1	Understand the different governing equations of theory of elasticity to solve for deformation field in a component.	K2
CO-2	Understand the methods of numerical differentiation and make proper selection based on the problem.	K2
CO-3	Understand the different methods of meshing and their advantages and drawbacks.	K2
CO-4	Understand the generation of stiffness matrix for simple problems.	K2
CO-5	Apply the concepts of planar and axial symmetry while using the FEM methods.	K3

UNIT-1 **(8 hours)**

Introduction to Finite Element Method**,** Element Calculations**,** Global Assembly**,** Application of Boundary Conditions and Higher Order Approximation, Review of various approximate methods variational approach and weighted residual approach application to structural mechanics problems, finite difference methods, governing equation and convergence criteria of finite element method.

UNIT-2 **(8 hours)**

Finite Element Method for Higher Order Approximation**,** Natural Coordinate**,** Numerical Integration**,** Programming of Element Calculation1-D Finite Element Code**,** bar elements, uniform section, mechanical and thermal loading, varying section, 2D and 3D truss element. Beam element - problems for various loadings and boundary conditions.

UNIT-3 **(8 hours)**

Pre-Processor**,** Processor and Post-Processor**,** Testing of 1-D FE Code**,** Integral Formulations for Beam Problem**,** Finite Element Formulation for Beam Problem: Shape Functions**,** Evaluation of Element Quantities and Assembly Procedure, Plane stress, plane strain and axi-symmetric problems, Derivation of element matrices for constant and linear strain triangular elements and axi-symmetric element.

UNIT-4 **(8 hours)**

Integral Formulations of Two-Dimensional Problems**,** Finite Element Formulation of 2-D Problems: FE Equations**,** Evaluation of Element Quantities, Assembly and Application of Dirichlet Boundary condition**,** Evaluation of Element Right Side Vectors, Assembly and Application of Dirichlet Boundary condition, stiffness matrix and consistent load vector, evaluation of element matrices using numerical integration.

UNIT-5 **(8 hours)**

Lagrangian Triangular Elements**,** Simplest Rectangular Element**,** Axi-symmetric Problems, Heat transfer problems, steady state fin problems, derivation of element matrices for two dimensional problems, torsion problems. Method of factorization for solving simultaneous algebraic equations – Features of software packages, sources of error, 3D Frame elements -

longitudinal and lateral vibration, Use of local and natural coordinates.

Books and Reference:

1. Tirupathi. R. Chandrapatha and Ashok D. Belegundu, "Introduction to Finite Elements in Engineering", Printice Hall India, Third Edition
2. Rao. S.S., "Finite Element Methods in Engineering," Butterworth and Heinemann
3. Reddy J.N., "An Introduction to Finite Element Method", McGraw Hill
4. Krishnamurthy, C.S., "Finite Element Analysis", Tata McGraw Hill
5. Bathe, K.J. and Wilson, E.L., "Numerical Methods in Finite Elements Analysis",
6. Larry J Segerlind, "Applied Finite Element Analysis", Second Edition, John Wiley and Sons.

Department Electives-V

Subject Code: KAE-075	Aircraft Rules and Regulations	L T P: 3 0 0	Credits: 3

After completion of this course, the students will be able to		Blooms Taxonomy
CO-1	Understand to work on aircraft according to the various aircraft acts, rules and responsibility of operator/owner.	K2
CO-2	Understand procedure for defect recording, reporting and investigation during the servicing of aircraft in AMO.	K2
CO-3	Apply the knowledge of maintenance to keep all the certificate updated in time to keep the aircraft airworthy.	K3
CO-4	Apply the knowledge for ageing aircraft to be in continue airworthy condition.	K3
CO-5	Apply knowledge of fuelling, defueling and precautions to taken while working on aircraft.	K3
CO-6	Understand operation and installation procedure of radio equipment's.	K2

UNIT-1 **(8 Hours)**

Rules and Regulation of Aircrafts: Need for Rules and Regulations as they relate to registration of aircraft in India. Aircraft Act 1934, Aircraft Rules 1937.

Procedure for issue of CAR and Responsibility of operator: Responsibility of operator/owners. Procedure for issue of CAR

Approval of cockpit check list, MEL & CDL: Minimum Equipment List Configuration Deviation List etc, Use of such documents. Preparation and use of cockpit and emergency check list.

UNIT-2 **(8 Hours)**

Defect Recording, Reporting, Monitoring and Investigation: Defect recording, reporting, investigation, rectification and analysis by the scheduled and other than scheduled operator. Fleet performance and preservation of records and component.

Approval of Organization: Requirements and procedure for issue/ extension in scope/renewal of organization in various categories: Layouts, Contents and requirements for quality control/ maintenance system/Quality Assurance Manuals.

UNIT-3 **(10 Hours)**

Airworthiness and Continued Airworthiness: General requirements for maintenance and certification of aircraft including gliders, micro-light aircraft, hot air balloons and rebuilding of Aircraft. Registration and deregistration of aircraft. Documents associated with continued airworthiness of aircraft such as certificate of airworthiness, issue revalidation and suspension of certificate of airworthiness, certificate of registration, Type certificate, flight

release, etc Procedures and requirements for issuance, renewal and restoration of validity of such documents, Conditions for suspension/cancellation of C of R, C of A, etc. Documents related with maintenance of aircraft and its components such as inspection schedules, special inspection schedules, TBO/COSL, Maintenance Planning Documents, manufacturers Literatures, Procedures for preparation and approval of such documents. Requirement for maintenance of aging aircraft, procedures and conditions for issuance of special flight permits.

UNIT-4 **(8 Hours)**

Requirement of Aircraft fuel, Refueling of Aircraft: Aircraft fuelling procedures. Un usable fuel supply-calibration of fuel quantity gauge of aircraft. Safety precautions against static electricity discharge bonding and earthing, fire hazard, hazards from adjacent aircraft operation. Aviation fuel at airport storage, handling and quality controls.

Aircraft instrument, equipment and accessories: General requirement for installation and maintenance of various instruments and systems installed on aircraft. Minimum instrument required and additional equipment as per requirement. Requirements for installation and maintenance of various mandatory equipment's on aircraft such as CVR, DFDR, ACAS, GPWS/ EGPWS and flight data recorders, combination recorders, data link recorders, airborne image recorders, airborne image recording system and aircraft data recording system.

UNIT-5 **(10 Hours)**

Airborne communication, navigation & Radar Equipment: Installation of communication, Navigation and Radar equipment. Installation of mode A, C and S transponder. Assignment of aircraft addresses. Procedure for allotment of mode S address.

Flight testing of aircraft: Flight Testing of aircraft for which a Certificate of Airworthiness has previously been issued.

Miscellaneous requirements: Weight and Balance Control of Aircraft. Provision of medical supplies in aircraft. Aircraft Log Books. Format of log books. Documents to be carried on board by Indian Registered aircraft and requirement. Procedure for issue of taxi permit, renewal of taxy permit.

Books and References:

1. Aircraft Act. 1934 & Aircraft rules 1937 by DGCA.
2. Civil Aviation Requirement Section-2 by DGCA.
3. Civil Aviation Requirement 21 by DGCA.
4. Civil Aviation Requirement 145 by DGCA.
5. Civil Aviation Requirement M by DGCA.
6. DGCA updates in www.dgca.nic.in

Subject Code: KAE-076	Aircraft Maintenance	L T P: 3 0 0	Credits: 3

After completion of this course, the students will be able to		Blooms Taxonomy
CO-1	Understand the construction, installation and operation of piston and gas turbine engine.	K2
CO-2	Understand correct alignment of piston and turbine engine in aircraft.	K2
CO-3	Understand the various controls of piston engine and gas turbine engine and monitoring in related instrument.	K2
CO-4	Apply the concept of periodic servicing and rectification of defects.	K3
CO-5	Understand propeller and their mechanism with integration to piston and turbine engine.	K3

UNIT-1 **(8 Hours)**

Introduction: Types of piston and gas turbine engines and their relative merits and demerits, principle of operation of piston and gas turbine engines, general constructional details and functions of each part, Principles, arrangement, operation and inspection of complete fuel, lubrication, ignition and staring system as applicable to piston and gas turbine engines, Principles and operation of accessories such as fuel pump, oil pump, barometric pressure control and air fuel ratio control etc. As applicable to gas turbine and piston engines.

UNIT-2 **(8 Hours)**

Knowledge of material used in piston and gas turbine engines. Methods of checking of components prior to and during subassembly and subsequent assembly of the engine of both piston and gas turbine type for correct alignment, Weight and balance Knowledge of various gauges and precision instruments used. Inspection procedures for crack detection Such as die-penitent, magnetic particle Inspection, X- Ray, ultra-sonic and eddy Current.

UNIT-3 **(8 Hours)**

Engine controls and performance instruments, General principles of operation and methods of checking for correct functioning. Testing of piston and gas turbine engines, various test rigs/instruments and equipment used. Fault diagnosis and rectification during testing. Engine installation checks of aircraft engines, precautions and procedures prior to, during and after ground running checks with particular reference to operating limitation, priming bleeding, Trouble shooting and rectification.

UNIT-4 **(8 Hours)**

Procedure for short term and long-term storage of engines. Accessories priming and inhibition procedures. Periodical servicing, procedures, reporting and rectification of defects and inspection after shock landings. Maintenance of inspection records and clearance procedure for flights. Types of stores, maintenance of stores. Inspection of documents and procedures for

storage of components.

UNIT-5 **(8 Hours)**

Types of propellers and their mechanisms, variable pitch, feathering propellers and associated control system components. Knowledge of lubricants, hydraulic fluids and fuels used in piston and gas turbine engines. Maintenance objectives & legal requirements as related to power plants & systems. Accident reporting & investigation. Power augmentation devices, thrust reversers & auxiliary power unit.

Books and References:

1. Jet Engine Manual by E. Mangham and APeace.
2. Fundamentals of Internal Combustion Engines by P.W. Gill, J.H. Smith & E.J.Ziurys.
3. Gas Turbine for Aircraft by A.W.Judge.
4. Gas Turbine Materials by G, Lueas and J.F.Pollock.

Subject Code: KAE-077	Introduction to CFD	L T P: 3 0 0	Credits: 3

After completion of this course, the students will be able to		Blooms Taxonomy
CO-1	Understand the governing equations of fluid flow in conservative and non-conservative forms.	K2
CO-2	Understand the methods of numerical differentiation.	K2
CO-3	Understand the different algorithms to capture shock in any flow field.	K2
CO-4	Apply the different methods of domain discretization to numerically solve any physical problem.	K3
CO-5	Apply boundary conditions corresponding to any CFD problem.	K3

UNIT-1 **(8 hours)**

Introduction to CFD: Principles of Conservation-Continuity Equation, Navier-Stokes Equation, Energy Equation and General Structure of Conservation Equations, Classification of Partial Differential Equations and Physical Behavior, Approximate Solutions of Differential Equations-Error Minimization Principles, Variation Principles and Weighted Residual Approach.

UNIT-2 **(8 hours)**

Finite Element Method, Finite Difference and Finite Volume Method, Some Conceptual Basics and Illustrations through 1-D Steady State Diffusion Problems, Boundary Condition Implementation and discretization of unsteady state Problems, Equations of fluid dynamics and their classification, Boundary conditions, Finite difference schemes- Projection and truncation error, Stability, consistency, accuracy and convergence of numerical schemes.

UNIT-3 **(8 hours)**

Basics of Finite Volume Method: Equations in integral form, numerical flux at cell faces, upwind methods, flux - vector splitting, flux- difference splitting, shock capturing methods, Introduction to CFD as a design tool; explicit and implicit methods; O.C.H types of grids. Important Consequences of discretization of time dependent diffusion type problems and stability analysis.

UNIT-4 **(8 hours)**

LAX Equivalence theorem, Stability analysis of parabolic equations (1-D unsteady state diffusion problems): FTCS (Forward time central space) scheme, Stability analysis of parabolic equations (1-D unsteady state diffusion problems): CTCS scheme (Leap frog scheme), Dunford-Frankel scheme, Stability analysis of hyperbolic equations: FTCS, FTFS, FTBS and CTCS Schemes, Finite Volume Discretization of 2-D unsteady State Diffusion type Problems.

UNIT-5 **(8 hours)**

Solution of Systems of Linear Algebraic Equations: Elimination Methods, Iterative Methods, Gradient Search Methods, Discretization of Convection-Diffusion Equations: A Finite Volume Approach, Discretization of Naiver Stokes Equations- Stream Function Vorticity approach and

Primitive variable approach, SIMPLE Algorithm, SIMPLER Algorithm, Unstructured Grid Formulation.

Books and References:

1. T.J.R. Hughes, The Finite Element Method: Linear Static and Dynamic Finite Element Analysis.
2. O.C. Zicnkievicz and R.L. Taylor, The Finite Element Method, Vol I & II, McGraw Hill
3. John. D. Anderson Jr., Computational Fluid Dynamics: The Basics with Applications, McGraw Hill.
4. Charles Hirsch, Numerical Computation of Internal and External Flows, Wiley Series I.

Subject Code: KAE-078	Elements of Aero elasticity	L T P: 3 0 0	Credits: 3

After completion of this course, the students will be able to		Blooms Taxonomy
CO-1	Understand the aerodynamic-structural interaction in an aircraft.	K2
CO-2	Understand the basics concepts of structural dynamics of a system of particles.	K2
CO-3	Understand the aero-elastic phenomenon like flutter and its effect on an aircraft.	K2
CO-4	Apply the concepts of vibrations and determine the modes and their frequencies for any system.	K3
CO-5	Analyze the structural strength of aircraft components with proper consideration of the effects of longitudinal and torsional vibrations.	K3

UNIT-1 **(8 hours)**

Rectilinear motion of a particle: Differential equation of motion in resisting medium, free vibrations with viscous damping, forced ration with harmonic disturbing force and general disturbing force, plane harmonic motion, Motion of a projectile with and without damp, motion of a particle subjected to a central force: planetary motion.

UNIT-2 **(8 hours)**

Dynamics of a System of Particles: Principle of Linear momentum and regular momentum, Rectilinear motion of a variable mass: Rockets, Kinetic energy and work, Law of conservation of energy. Equations of constraints, generalized coordinates, generalized forces, Equations of equilibrium, Generalized co-ordinates, Application of generalized coordinates in bending of beams, D' Alembert's principle, Lagrange's equation and applications, Hamilton's principle and application.

UNIT-3 **(8 hours)**

Small Oscillations of Conservative Systems: Free vibrations of derivative systems, Linear oscillations of two coupled masses, Free variation of system with two degrees of freedom and system with several degrees of freedom, Principal modes and their orthogonal property, Normal modes static coupling and dynamic coupling, Approximate methods of calculating principal frequencies.

UNIT-4 **(8 hours)**

Dynamics of Elastic Bodies: Vibration of a string under tension, Free vibration of beams with various end condition and the determination of the various modes of vibrations and their natural frequencies, variation of beams with concentrated masses, Critical speed of a rotating Forced vibration of beams, Tensional vibration of a shaft and –shaft combination, Approximate methods of calculating natural frequencies.

UNIT-5 **(8 hours)**

Nature of static aero elastic phenomenon , Wing diver, Gene and control system reversal for an idealized tow dimensional wing and approximate solution for a finite wing , Flutter phenomena

and flutter analysis, Difference between flutter instability and resonance, Simplified expressions for aerodynamic forced and moments for an oscillating air foil, Determination of flutter speed and frequency for an idealized two dimensional wing as well as for a finite wing, Methods of flutter control and prevention, Elementary theory of buffeting.

Books and References:

1. W.T. Thomson, Vibration Theory and Application, Allen and Unwind
2. S. Timoshenko and D.H, Young, Advanced Dynamics , McGraw Hill
3. Y.C. Fung, Introduction to the Theory of Aero elasticity, Addison Wesley